Dynamic Execution Commitment of Vision-Language-Action Models

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Abstract

Vision-Language-Action (VLA) models predominantly adopt **action chunking**, *i.e.*, predicting and committing to a short horizon of consecutive low-level actions in a single forward pass, to amortise the inference cost of large-scale backbones and reduce per-step latency. However, committing these multi-step predictions to real-world execution requires balancing success rate against inference efficiency — a decision typically governed by fixed execution horizons tuned per task. Such heuristics ignore the state-dependent nature of predictive reliability, leading to brittle performance in dynamic or out-of-distribution settings. In this paper, we introduce A^3, an Adaptive Action Acceptance mechanism that reframes dynamic execution commitment as a self-speculative prefix verification problem. A^3 first computes a trajectory-wise consensus score of actions via group sampling, and then selects a representative draft and prioritizes downstream verification. Specifically, it enforces: (1) consensus-ordered conditional invariance, which validates low-consensus actions by judging whether they remain consistent when re-decoded conditioned on high-consensus actions; and (2) prefix-closed sequential consistency, which guarantees physical rollout integrity by accepting only the longest continuous sequence of verified actions starting from the beginning. Consequently, the execution horizon emerges as the longest verifiable prefix satisfying both internal model logic and sequential execution constraints. Experiments across diverse VLA models and benchmarks demonstrate that A^3 eliminates the need for manual horizon tuning while achieving a superior trade-off between execution robustness and inference throughput.

1 Introduction

Vision-Language-Action (VLA) models [1, 2, 3] have emerged as a primary paradigm for generalizable embodied intelligence [4, 5], mapping high-dimensional visual observations and natural language instructions directly to precise motor sequences. To mitigate the high computational overhead of large-scale vision-language backbones [6], modern VLA architectures increasingly adopt dual-system designs [1] that decouple deliberative reasoning from reactive execution. A core efficiency strategy within these frameworks is action chunking, where an action expert (e.g., π-0.5 [1] or GR00T [7]) generates a sequence of future actions in a single forward pass before interacting with the environment again. While substantially reducing inference frequency and improving computational utilization, the determination of the execution horizon, *i.e.,* the number of actions actually committed before the next re-planning cycle, remains a critical yet underexplored design choice.

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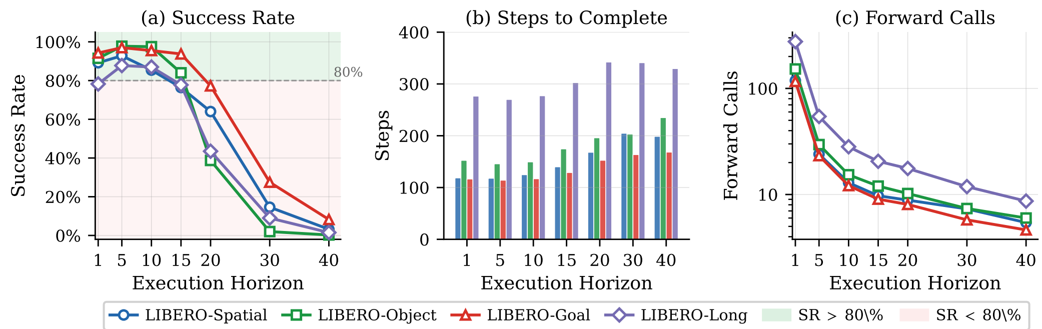


Figure 1: Performance analysis of π-0.5 under varying execution horizons on LIBERO benchmark [8]. (a) Success rate first increases while then decreases as horizon increases, dropping below 80% when horizon is larger than 15 for most suites. (b) Completion step increases substantially with a larger horizon, as failed recoveries and compounding errors drive total steps up to $1.5\times$ higher than at horizon=1. (c) Forward calls decrease monotonically with a longer horizon, revealing a fundamental trade-off between inference overhead and success rate.

In existing VLA systems, execution commitment is typically restricted to a predefined, static horizon selected a priori per task [9]. This design implicitly assumes that the model's predictive reliability is uniform across the temporal rollout and invariant to environmental stochasticity. However, in open-ended embodied settings, observations are non-stationary and execution errors inevitably compound, rendering a fixed execution horizon substantially brittle [10, 11]. As illustrated in Figure 1, it reveals a fundamental trade-off between inference efficiency and task success rate under varying execution horizons: a smaller execution horizon requires more frequent model queries, imposing heavy inference overhead on real-time robot control, whereas a larger horizon reduces forward calls but commits to longer open-loop action sequences, resulting in severe success rate degradation as accumulated errors cannot be corrected by visual feedback.

This motivates the design of an adaptive mechanism that reduces unnecessary forward calls while preserving the reactivity needed to maintain high task success. While recent efforts have begun exploring adaptive execution, existing approaches can be broadly categorized into two paradigms, each with distinct limitations. *Consistency-based methods*, such as Mixture of Horizons (MoH) [10] and EveryDayVLA [11], estimate reliability by measuring agreement across multiple prediction sources—either cross-horizon segment consensus or discrepancy between discrete and continuous action heads. However, these approaches either require additional training overhead or produce heuristic thresholds that are neither calibrated nor grounded in sequential execution feasibility. *Attention-based methods*, such as AutoHorizon [12], leverage self-attention weights as a proxy [13, 14] for predictive confidence, but attention patterns do not directly reflect the physical feasibility of committing an action sequence in an open-loop manner. As a result, none of these methods provides a principled mechanism that jointly accounts for model-level self-consistency and execution-level sequential feasibility in continuous action spaces.

To resolve this tension, we introduce Adaptive Action Acceptance (A^3), a mechanism that reframes dynamic execution commitment as a self-speculative prefix verification problem. Inspired by the principles of self-speculative decoding [15, 16], A^3 treats the predicted action sequence as a draft and adaptively determines the trustworthy execution horizon by assessing the model's internal logic consistency and sequential execution consistency, without auxiliary modules or retraining. We first compute a mode-aware trajectory consensus score of actions over multiple sampled rollouts. The goal of consensus scoring is to estimate, for each action, how stably the model predicts that action across independent rollouts - not in raw action space (where incremental errors compound), but in the space of induced trajectory states. This score serves as a self-consistency prior for representative draft selection and verification ordering, rather than a calibrated estimate of correctness. Then A^3 determines the execution window through a dual hierarchical verification pipeline. First, it enforces *consensus-ordered conditional invariance*: lower-consensus actions are re-decoded conditioned on higher-consensus actions fixed as context; an action is accepted only if it remains invariant under this re-decoding, ensuring the speculative segment is grounded in the model's committed long-term plan.

Second, it imposes *prefix-closed sequential consistency*: each action is re-decoded conditioned on all its temporally preceding verified actions; even if an action passes consensus-ordered conditional invariance, it is accepted only if this sequential re-decoding also confirms invariance, ensuring the committed prefix forms an unbroken causal chain robust to covariate shift. The final execution horizon emerges as the longest prefix satisfying both constraints, resolved in a single parallel forward pass—adapting naturally to task difficulty and observation quality without any task-specific tuning. Our contributions can be summarized as:

• We identify execution commitment as a missing inference principle in multi-step VLA and reformulate horizon selection as a state-conditioned prefix verification problem.

• We propose A^3, a dual hierarchical verification framework that enforces consensus-ordered conditional invariance and prefix-closed sequential consistency, enabling execution horizons to emerge from structured validation rather than predefined heuristics.

• We demonstrate across multiple VLA benchmarks that A^3 removes the need for task-specific horizon tuning and exposes a controllable trade-off between execution reliability and inference efficiency.

2 Related Work

Vision-language-action model. VLA models [17, 18] aim to learn policies that map visual observations and language instructions [19, 20] directly to continuous control actions [21]. Early approaches [2] primarily relied on autoregressive or behavior-cloning policies that predict actions step by step. More recently, dual-system VLA architectures [1, 9] have emerged, decoupling high-level multimodal reasoning [19, 22] from low-level action generation. In these systems, a perception-language backbone [19] produces contextual representations, while a dedicated action expert is often implemented as a diffusion [23] or flow-matching head that predicts multiple future actions in parallel [24, 25, 26]. In practice, these systems often execute a predefined number of predicted actions before re-planning, with the execution horizon typically selected per task or benchmark [27, 28, 29]. For instance, π-0.5 [1] adopts different execution horizons across LIBERO subtasks [8] to balance performance and stability. Although effective under benchmark-specific settings, such task-dependent configuration relies on prior knowledge of task structure.

Speculative decoding. Speculative decoding [16, 30] was originally proposed to accelerate autoregressive language models by drafting multiple tokens and verifying them using a stronger or identical model. Subsequent work, including self-speculative decoding [15, 31], further improves efficiency by leveraging structured verification trees to enable parallel token prediction while preserving exact decoding correctness in discrete token spaces. For instance, SSD [15] leverages self-drafting to eliminate auxiliary models, enabling dLLMs [32] to generate and verify multiple tokens per iteration [33]. While structurally related [34], our work addresses a fundamentally different objective. Speculative decoding aims to improve generation efficiency without altering model outputs. In contrast, we focus on execution commitment in VLA systems—determining how much of a predicted continuous action sequence can be safely executed under the current state. Instead of enforcing exact token matching, our framework models commitment as a consensus-ordered, prefix-closed verification process that explicitly accounts for uncertainty in continuous action spaces and sequential feasibility in embodied environments.

Adaptive execution and horizon selection. Prior work has explored adaptive horizons primarily through model-based control (MPC) [35] or confidence-based heuristics [36, 35]. In the VLA context, EveryDayVLA [11] utilizes the AdaHorizon algorithm to adjust windows based on disagreement between discrete and continuous action heads. Similarly, NanoVLA [37] employs Long-Short Action Chunking to plan long but act short. Mixture-of-Horizon (MoH) [10] rearranges action chunks into multiple segments with different horizons and fuses their predictions through a lightweight linear gating mechanism. Moreover, AutoHorizon [12] depends on the attention weight as prediction confidence to infer the temporal limit of the model's reliable forecasting capability. In contrast, A^3 does not rely on explicit dynamics models, auxiliary heads, or passive thresholds. Instead, it treats commitment as a self-consistent prefix verification problem where the horizon emerges from conditional invariance, providing a principled way to scale execution without task-specific tuning.

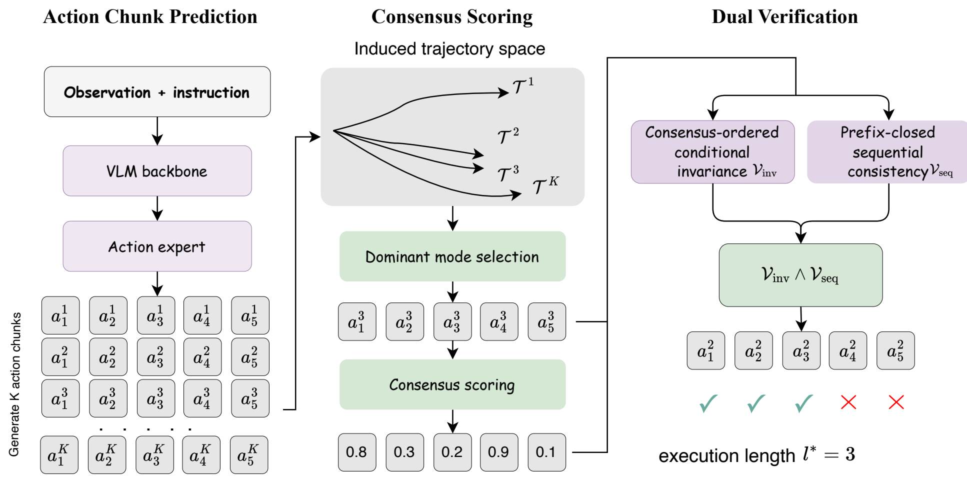


Figure 2: Overview of A^3. Given the current observation and instruction, the VLM backbone and action expert generate K candidate action chunks. The chunks are mapped to induced trajectory states, from which the dominant mode is identified via clustering and its medoid selected as the primary draft; per-step consensus scores reflect the model's self-consistency at each action position. The draft then undergoes dual parallel verification in a single batched forward pass, yielding the execution horizon l^* as the longest verified prefix. We also illustrate the implementation of dual verification tree in Figure 6.

3 Method

As illustrated in Figure 2, the A^3 framework transforms static action commitment into an adaptive, state-conditioned verification process via a three-stage pipeline. First, A^3 generates K candidate chunks in a single forward pass and then computes a mode-aware trajectory consensus score to select a primary draft sequence. The draft is then verified through a parallel tree governed by two hierarchical constraints: (1) *Consensus-ordered Conditional Invariance*, which validates lower-consensus actions by judging whether they remain invariant when re-decoded conditioned on higher-consensus actions; and (2) *Prefix-closed Sequential Consistency*, which enforces physical causal integrity by accepting an action only if all its predecessors satisfy verification. Consequently, the final execution horizon l^* emerges as the longest prefix satisfying both model logic self-consistency and physical rollout constraints.

3.1 Execution Commitment Formulation

Consider a VLA policy π_θ that, given the current state s_t (including visual observations and language instructions), predicts a sequence of future actions of length H, written as

$$\mathbf{A}_t = \pi_\theta(s_t) = \{a_t, a_{t+1}, \dots, a_{t+H-1}\}. \tag{1}$$

Due to the non-stationarity of embodied environments and the progressive accumulation of prediction errors, executing the full action sequence $\mathbf{A}_t$ in an open-loop manner is generally suboptimal. Instead, the system should adaptively determine an appropriate execution horizon l to balance success rate against inference efficiency, as illustrated in Figure 1. We formalize execution commitment by selecting a prefix length $l \le H$ and defining the committed action subsequence as

$$\mathbf{A}_t[1:l] = \{a_t, \dots, a_{t+l-1}\}, \tag{2}$$

which can be safely executed under the current state s_t. Formally, we determine the prefix length by solving the following constrained optimization problem:

$$l^* = \underset{l \in \{1,\dots,H\}}{\arg\max}\ l \quad \text{s.t.} \quad \prod_{\tau=t}^{t+l-1} \mathcal{V}(a_\tau, s_\tau) = 1. \tag{3}$$

Here, $\mathcal{V}(a_\tau, s_\tau) \in \{0, 1\}$ is a verification function that indicates whether the candidate action prefix satisfies the execution constraints under the current state s_t.

3.2 Mode-aware Trajectory Consensus Scoring

The primary challenge in assessing the consistency of an action sequence $\mathbf{A}_t$ is the intra-sequence visual open-loop vulnerability. Since VLA models typically output incremental displacements (e.g., Δ-poses), small per-step prediction errors accumulate through temporal integration, leading to significant task-space drift. Traditional uncertainty metrics that focus on marginal action variance at each timestep [11] fail to capture this compounding physical reality.

Induced trajectory states. To account for error propagation, A^3 evaluates consistency in the space of induced trajectory states. We define the induced state at a future step $t + j$ as:

$$\hat{s}_{t+j} = \mathcal{G}(s_t, \{a_t, \dots, a_{t+j-1}\}), \tag{4}$$

where $\mathcal{G}$ denotes kinematic integration [38] of incremental actions into a cumulative pose. A^3 probes the model's posterior by generating K independent candidate sequences $\{\mathbf{A}_t^{(1)}, \dots, \mathbf{A}_t^{(K)}\}$ in a single forward pass, yielding a set of induced trajectories $\{\mathcal{T}^{(1)}, \dots, \mathcal{T}^{(K)}\}$.

Dominant mode selection. Rather than measuring consensus against a global centroid, which conflates multi-modal but internally coherent futures with genuine instability, we identify the dominant trajectory mode via clustering. Since spatially similar rollouts may be temporally phase-shifted, we measure pairwise trajectory agreement using a locally time-aligned distance $d_j^{(k,k')} = \min_{|\Delta| \le w} \|\hat{s}_{t+j}^{(k)} - \hat{s}_{t+j+\Delta}^{(k')}\|_2^2$, aggregated over steps as $D(k, k') = \sum_j d_j^{(k,k')}$. We cluster the K trajectories under this distance and score each cluster c by

$$S_c = p_c \cdot \exp(-\mathrm{Disp}(c)/\tau), \tag{5}$$

where $p_c = |c|/K$ is the cluster mass, $\mathrm{Disp}(c)$ is the mean pairwise distance within the cluster, and $\tau > 0$ is a temperature parameter controlling the compactness penalty. The dominant mode is then selected as $c^* = \arg\max_c S_c$, balancing trajectory coherence with population support.

Consensus score of action. Within dominant mode c^*, we select the medoid trajectory $k^* = \arg\min_{k \in c^*} \sum_{k' \in c^*} D(k, k')$ as the primary draft, and define the action consensus score as:

$$\tilde{R}_j = p_{c^*} \cdot \exp\left(-\frac{1}{|c^*|} \sum_{k \in c^*} d_j^{(k,k^*)}\right), \tag{6}$$

where $d_j^{(k,k^*)}$ is the time-aligned distance from sample k to the medoid at step j. Higher $\tilde{R}_j$ indicates that rollouts concentrate around a stable dominant plan at that step.

Importantly, we do not interpret $\tilde{R}_j$ as a calibrated measure of correctness. Rather, it serves as a *self-consistency prior*: it selects the primary draft and determines the order in which actions are subjected to downstream verification. Final execution commitment is decided by the subsequent conditional-invariance and prefix-consistency checks, rather than by consensus alone.

3.3 Dual Hierarchical Verification

Following the drafting phase, A^3 adjudicates the draft's trustworthiness through a formal verification phase. Unlike token-level matching in LLMs, our framework treats verification as a progressive anchoring process governed by two constraints designed for embodied consistency.

Consensus-ordered conditional invariance. This constraint evaluates whether a speculative action remains stable when the model's most certain predictions are fixed as context. For each candidate action a_{t+i}, we define a consensus context $\mathcal{C}_i^{\mathrm{inv}}$ consisting of all actions with higher consensus scores:

$$\mathcal{C}_i^{\mathrm{inv}} = \{a_{t+j} \mid \tilde{R}_j > \tilde{R}_i,\ j \in \{1, \dots, H\}\}, \tag{7}$$

and verify a_{t+i} by re-decoding it conditioned on this context:

$$\mathcal{V}_{\mathrm{inv}}(a_{t+i}, s_t \mid \mathcal{C}_i^{\mathrm{inv}}) = \mathbb{I}\left(\left|\pi_\theta(s_t \mid \mathcal{C}_i^{\mathrm{inv}})[i] - a_{t+i}\right| < \delta\right), \tag{8}$$

where $\pi_\theta(s_t \mid \mathcal{C}_i^{\text{inv}})[i]$ is the re-decoded output at position i, and $\delta > 0$ is an acceptance threshold. An action is accepted only if the model re-confirms it after its higher-consensus counterparts are locked in, ensuring that acceptance is grounded in the model's internal logic rather than marginal prediction alone.

Prefix-closed sequential consistency. Robotic execution is a strictly causal and non-reversible process. A failure at action j invalidates the environmental assumptions for all subsequent actions, as the resulting state would diverge significantly from the model's training distribution—a phenomenon known as covariate shift. To prevent the robot from skipping across unverified segments, we enforce prefix-closed sequential consistency. Under this constraint, an action i is only valid if all preceding actions in the chain are treated as fixed context and the action remains stable under re-decoding. For each action i, we define a sequential prefix context consisting of all preceding actions:

$$\mathcal{C}_i^{\text{seq}} = \{a_{t+j} \mid j < i\}. \tag{9}$$

We then perform an analogous re-decoding procedure for a_{t+i} under the sequential context and verify it as:

$$\mathcal{V}_{\text{seq}}(a_{t+i}, s_t \mid \mathcal{C}_i^{\text{seq}}) = \mathbb{I}\left(|\pi_\theta(s_t \mid \mathcal{C}_i^{\text{seq}})[i] - a_{t+i}| < \delta\right). \tag{10}$$

3.4 Emergent Horizon Determination

The final execution commitment l^* emerges as the global solution to the following constrained maximization problem:

$$\begin{aligned} l^* = &\arg\max_{l \in \{1,\dots,H\}} \; l \\ \text{s.t.} \quad &\mathcal{V}_{\text{inv}}(a_{t+l}, s_t \mid \mathcal{C}_l^{\text{inv}}) = 1, \quad \mathcal{V}_{\text{seq}}(a_{t+l}, s_t \mid \mathcal{C}_l^{\text{seq}}) = 1. \end{aligned} \tag{11}$$

To maintain real-time throughput, A^3 implements this progressive verification as a hierarchical tree over candidate prefixes. By exploiting the parallel processing of the action expert, the entire dual-stage verification for the prediction horizon H is completed in a single parallel forward pass. More implementation details and the algorithm can be found in Sec A.

4 Experiments

4.1 Experiential Setup

Implementation details. We implement A^3 on three VLA models including π_0 [9], $\pi_{0.5}$ [1], and GR00T [7]. All model configurations are adopted from the RLinf codebase [39]. The prediction horizon H follows each backbone's default setting on the respective datasets. Specifically, we set $H = 10$ for LIBERO [8], $H = 8$ for ManiSkill [40], and $H = 5$ for MetaWorld [41] when using π_0 and $\pi_{0.5}$, whereas $H = 5$ is adopted for GR00T. During group probing, we generate $K = 8$ candidate chunks via parallel noise sampling in a single batched forward pass. For dual verification, considering the varying action dimensions across datasets (7 for LIBERO and ManiSkill, and 4 for MetaWorld), the invariance constraint is applied independently on each continuous action dimension, excluding the final binary gripper state. The threshold δ defaults to the dimension-specific standard deviation used for action denormalization in the original model checkpoint of the corresponding dataset. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

Benchmarks. We evaluate A^3 on three representative embodied manipulation benchmarks: MainSkill [40], MetaWorld [41], and LIBERO [8]. These benchmarks cover diverse multi-stage tasks requiring long-horizon planning, precise continuous control, and language-conditioned execution, providing a comprehensive testbed for adaptive execution commitment. We additionally conduct real-robot experiments to validate the practicality of A^3, including four tasks: *FlipMug*, *TapeBox*, *HangMug*, and *StackCube*. Specifically, we consider a tabletop manipulation setting equipped with an Agilex Piper robotic arm, where two Intel RealSense D435i cameras are deployed as the primary and wrist observations. Both A^3 and the baselines are instantiated using π-0.5, which is trained on a dataset that consists of four tasks, each with 50 expert demonstrations collected via teleoperation. During training, the horizon is set to $H = 20$, and the control frequency of the robotic arm is fixed at 30 Hz. For evaluation, each result is obtained from 20 rollouts, except for the StackCube task, which is evaluated with 30 rollouts.

Table 1: Dynamic horizon comparison across LIBERO, MetaWorld, and MainSkill. We report average task success rate (Avg.) and average committed horizon length (Len.).

Backbone	Method	LIBERO						MetaWorld		MainSkill	
		Spatial	Object	Goal	Long	Avg.	Len.	Avg.	Len.	Avg.	Len.
π-0	Original	97.2	98.2	95.6	84.6	95.1	5	78.0	3	77.6	5
	MoH	97.6	98.8	**96.4**	**87.4**	95.1	5	**79.4**	3	78.0	5
	A^3 (Ours)	**98.2**	**99.2**	96.1	87.2	**95.3**	**9.4**	79.2	**3.2**	**78.6**	**6.2**
π-0.5	Original	98.6	**99.8**	97.6	95.4	97.9	6.3	77.8	3	88.1	5
	MoH	98.4	99.0	**98.2**	95.2	97.7	5	78.4	3.3	88.4	5
	EverydayVLA	98.2	99.2	97.4	95.6	97.6	6.8	-	-	-	-
	AutoHorizon	**99.1**	99.2	97.5	91.6	96.9	-	-	-	-	-
	A^3 (Ours)	98.6	99.4	98.0	**96.2**	**98.1**	**9.7**	**79.4**	**4.5**	**89.1**	**5.2**
GR00T	Original	91.4	97.8	84.4	86.8	90.1	4.7	-	-	-	-
	A^3 (Ours)	**93.8**	**99.2**	**87.2**	**91.4**	**92.9**	4.5	-	-	-	-

Table 2: Performance comparison on real-world tasks.

Setting	Exec Horizon	Success Rate (%)					Inference Calls	Total Steps
		FlipMug	TapeBox	HangMug	StackCube	Average		
Fixed horizon	5	50	35	25	0	27.5	91.5	455.3
	10	70	**100**	35	66.7	67.9	32.7	321.5
	15	95	**100**	35	**86.7**	79.2	17.7	257.7
	20	90	95	35	73.3	73.3	12.4	238.1
Ours	13.5	**95**	**100**	**60**	83.3	**84.6**	17.2	250.1

Metrics. We report **task success rate** as the primary performance metric. To characterize execution behavior, we report the **average execution horizon length**, defined as the number of actions committed before each replan. This metric reflects how aggressively or conservatively a method commits predicted action chunks. To measure computational cost, we report the **average number of replans**, defined as the total number of VLA forward passes invoked during an episode.

Baselines. We compare A^3 against three representative execution strategies. **Fixed Horizon.** This baseline reflects the standard execution paradigm for VLA models, where a fixed number of actions $l \in \{1, \ldots, H\}$ from each predicted chunk are executed before re-observing the environment. To provide a strong baseline, we determine the optimal fixed horizon via exhaustive grid search on the evaluation suite, selecting the l that maximizes task success rate. **Mixture of Horizons (MoH).** MoH [10] mitigates the trade-off between long-horizon foresight and short-term accuracy by decomposing the action chunk into segments with different effective horizons. Predictions from multiple horizon-specific segments are fused using a lightweight linear gating mechanism. Execution commitment is determined through cross-horizon consensus, forming a self-truncated executable chunk based on agreement among the different horizon voters. **Adaptive Horizon.** We additionally compare with state-of-the-art uncertainty-based adaptive execution methods. A representative approach is AdaHorizon of EveryDayVLA [11], which estimates model uncertainty by measuring the discrepancy between concurrently generated continuous and discrete action predictions. Besides, AutoHorizon [12] leverages self-attention weights as a proxy for the model's predictive limit.

4.2 Main Results

Comparison with baselines. Table 1 shows that A^3 consistently maintains or improves task success while substantially extending the committed horizon. On π-0, A^3 matches the best LIBERO average success rate (95.1%) but increases the execution length from 5 to 9.4 steps, and improves MainSkill from 77.6% to 78.6%. On the stronger π-0.5, A^3 achieves the highest LIBERO average success (98.1%) while extending the committed horizon from 6.3 to 9.7 steps, a 54% increase in execution length with simultaneous performance gains. Meanwhile, our method achieves better performance efficiency tradeoff on Metaworld with 1.6% improvement in success rate and future 1.5 horizon

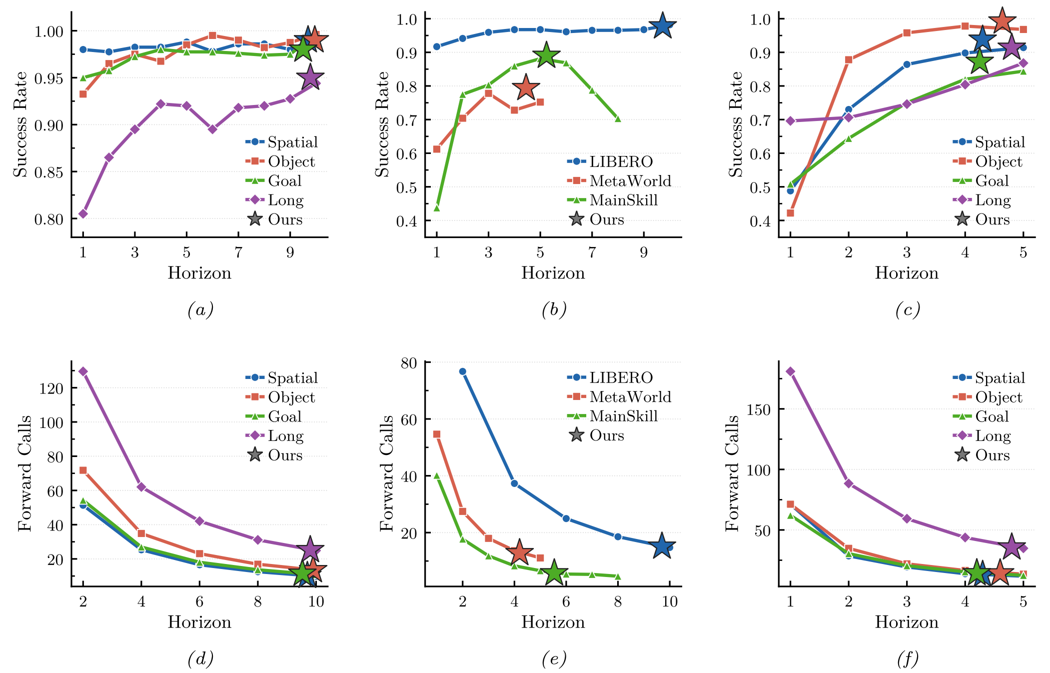


Figure 3: Trade-off between success rate (top row) and forward calls (bottom row) across execution horizons. The first two column experiments are conducted on π-0.5 while the last one is on GR00T.

extension than manual choosing. These results demonstrate that our method enables longer yet reliable commitment, with execution horizons emerging adaptively from task difficulty.

Efficiency–performance tradeoff. Figure 3 illustrates the tradeoff between execution horizon, success rate, and inference calls across task suites and benchmarks. Longer horizons generally reduce forward calls but risk error accumulation, leading to saturated or degraded success rates on challenging tasks. Our method adaptively selects the execution horizon, placing itself near the Pareto frontier: it achieves competitive or superior success rates while requiring significantly fewer forward calls compared to fixed short-horizon baselines, demonstrating that adaptive commitment effectively balances performance and inference efficiency.

Real-world results. Table 2 summarizes the performance of A^3 on four physical manipulation tasks. A^3 achieves an average success rate of 84.6% across FlipMug, TapeBox, HangMug, and StackCube, outperforming the best fixed-horizon baseline (79.2% at horizon 15) while maintaining comparable inference calls (17.2 vs. 17.7) and total steps (250.1 vs. 257.7). The adaptive execution horizon converges to 13.5 steps on average without any task-specific tuning. Notably, the gains are most pronounced on contact-rich tasks: HangMug improves by 25% over the best fixed horizon, as the adaptive mechanism shortens the committed prefix during the sub-centimeter alignment phase to maintain tighter visual feedback. In contrast, TapeBox achieves 100% under both settings, suggesting that the task's relatively forgiving placement tolerance does not require fine-grained horizon adaptation. StackCube, which involves precise stacking under significant positional uncertainty, benefits most from shorter committed horizons during the critical placement phase, yielding a 16.6% gain. These results demonstrate that A^3 naturally allocates control granularity according to task difficulty—committing longer horizons during free-space motion and shortening them at contact-critical moments—without any task-specific configuration.

4.3 Robustness Analysis

State perturbations. We evaluate robustness under degraded observations by applying test-time *Masking* (occluding 50/55/60% of pixels) and *Blur* (Gaussian blur with kernel size (k=11/13/15));

Table 3: Performance comparison under different state perturbations using π-0.5 on LIBERO. Masking indicates the fraction of pixels occluded. Blur uses Gaussian blur with kernel size k.

Method	Original		Masking						Blur					
	Avg ↑	Len	50%		55%		60%		k=11		k=13		k=15	
			Avg ↑	Len	Avg ↑	Len	Avg ↑	Len	Avg ↑	Len	Avg ↑	Len	Avg ↑	Len
Original	97.9	6.3	83.2	6.3	66.6	6.3	40.8	6.3	92.6	6.3	79.6	6.3	55.8	6.3
A^3 (Ours)	98.1	9.7	89.0	8.9	72.6	7.5	51.0	6.3	96.8	9.6	89.6	8.9	66.0	8.1
	+0.2	+3.4	+5.8	+2.6	+6.0	+1.2	+10.2	0	+4.2	+3.3	+10.0	+2.6	+10.2	+1.8

Table 4: Effect of prediction horizon on execution horizon and LIBERO suites.

Pred. Horizon	**Method**	**Exec. Horizon**	**Spatial**	**Object**	**Goal**	**Long**
10	Fixed horizon	10	98.6	99.6	97.6	94.6
	Ours	9.7	98.6	99.4	98.0	96.2
15	Fixed horizon	15	91.4	96.4	96.4	89.6
	Ours	14.2	97.2	98.6	96.8	92.6
20	Fixed horizon	20	64.4	79.2	82.6	71.8
	Ours	15.9	72.0	81.4	84.2	73.6

results are summarized in Table 3. A^3 consistently improves success rate over the fixed-horizon baseline across all perturbations, with larger gains under stronger corruption (e.g., 10.2% improvement under k=15 Gaussian blur). Meanwhile, A^3 generally achieves longer execution horizons, indicating more stable commitment under noisy states. Under the hardest 60% masking, the committed horizon converges to 6.3 while still substantially increasing 10.2% success rate. Overall, these results demonstrate that A^3 effectively mitigates error accumulation and remains reliable under realistic state degradations.

Effect of prediction chunk size. Table 4 examines how prediction chunk size affects execution horizon and task performance on LIBERO suites. At a chunk size of 10, both methods perform comparably, with our method showing a marginal improvement on Long tasks. As the chunk size increases, the fixed-horizon baseline degrades substantially, most notably on Spatial, where the success rate drops from 98.6 to 64.4. Our method consistently mitigates this degradation by adaptively reducing the committed execution horizon: at chunk sizes of 15 and 20, we outperform the fixed baseline across all four task suites, with the largest gains on Spatial (5.8% and 7.6% respectively).

5 Ablation Study

Effect of components. Table 5 presents an ablation study of our key components on GR00T-LIBERO, π0.5-MainSkill, and π0-MetaWorld. Starting from the baseline, adding *consensus scoring* yields a small but consistent improvement in success for the draft selection while keeping the horizon unchanged, indicating that drafting alone mainly benefits local action quality. Introducing *consensus-ordered conditional invariance* brings the largest gain in success across all benchmarks (e.g., 90.6→92.1 on GR00T-LIBERO and 78.2→78.6 on π0-MetaWorld), and noticeably increases the executed horizon on MainSkill and MetaWorld, suggesting that filtering low-consensus segments improves both stability and commitment length. Finally, *prefix-closed sequential consistency* further boosts performance (92.1→92.9; 88.3→89.1; 78.6→79.2), demonstrating that enforcing temporal consistency reduces error accumulation over longer rollouts. Overall, these results show that each module contributes, with verification playing a central role and sequential checks providing complementary gains.

Parameter sensitivity analysis. We analyze the two key hyperparameters of A^3 on π-0.5 under LIBERO in Table 6. For the sample count K, success rate saturates quickly, with $K = 8$ recovering most gains over $K = 4$ while adding only 18.4ms overhead; we adopt $K = 8$ as default. For the acceptance threshold δ, both success rate and average horizon remain stable across $\delta \in \{0.5, 1.0, 1.5\} \times$ *std*, confirming that setting δ to the dimension-specific denormalization standard deviation provides a natural, dataset-adaptive calibration without manual tuning.

Table 5: Ablation study of each component on three benchmarks. We report the success rate/execution horizon length for each setting.

Module	GR00T-LIBERO	π0.5-MainSkill	π0-MetaWorld
Baseline (original)	90.1/4.70	88.1/5.00	78.0/3.00
+ Consensus scoring	90.6/4.70	88.1/5.00	78.2/3.00
+ Consensus-ordered conditional invariance	92.1/4.74	88.3/5.77	78.6/3.41
+ Prefix-closed sequential consistency	92.9/4.52	89.1/5.24	79.2/3.23

Table 6: Parameter sensitivity of K and δ on π-0.5 under LIBERO. Here, "std" denotes the dimension-specific standard deviation used for action denormalization in the original model checkpoint of the corresponding dataset.

parameter	success rate	latency(ms)	parameter	success rate	average horizon
K = 4	97.3	467.2	δ = 0.5 * std	97.9	9.6
K = 8	98.1	485.6	δ = 1 * std	98.1	9.7
K = 16	98.2	503.6	δ = 1.5 * std	97.8	9.8

Latency breakdown. Table 7 decomposes the per-step wall-clock latency of A^3 against the original fixed-horizon baseline. VLM observation encoding remains unchanged (70.1ms), confirming that A^3 introduces no overhead to the perception backbone. The additional cost comes from two sources: chunk generation increases from 146.1ms to 204.1ms due to the K-sample batched forward pass, and verification adds 211.5ms for the dual hierarchical re-decoding. Critically, this per-step overhead is amortized over a substantially longer committed horizon (9.7 vs. 6.3 steps), resulting in fewer total forward calls per episode and a net reduction in episode-level inference cost.

Table 7: Inference latency breakdown (per environment step). We decompose wall-clock latency (ms) into (i) VLM observation encoding, (ii) chunk generation, and (iii) verification.

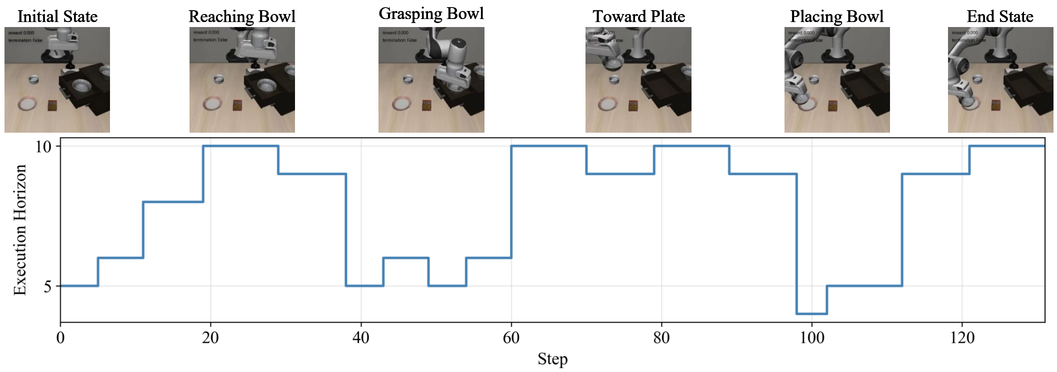
Method	Total	VLM obs.	Chunk gen.	Verification
Original	216.2	70.1	146.1	–
A^3 (ours)	485.6	70.0	204.1	211.5

Visualization of adaptive horizon behavior. Figure 4 visualizes how the execution horizon evolves throughout the rollout on three representative tasks. A consistent pattern emerges across all of them. During free-space motion, such as approaching the target object or transferring between locations, the policy adopts a relatively large horizon to accelerate execution. In contrast, during contact-rich phases such as grasping, placing, and flipping, the horizon automatically shrinks to ensure sufficient feedback frequency for fine-grained control.

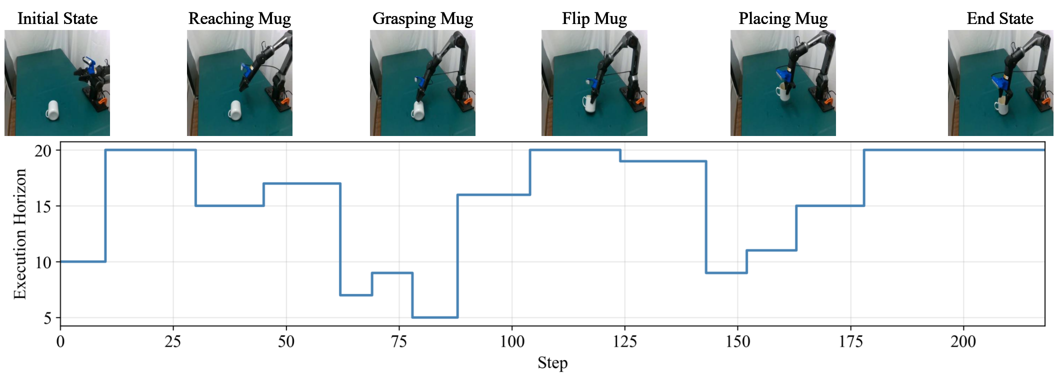
For the task of picking up the black bowl from the top drawer of the wooden cabinet and placing it on the plate, shown in Fig. 4(a), the rollout involves reaching into a confined space, grasping the bowl, transferring it out of the drawer, and finally placing it on the plate, leading to frequent alternation between long and short horizon segments. The flip-the-mug task in Fig. 4(b) requires the gripper to grasp the rim of an inverted mug, lift and rotate it upright, and then release it, where free-space transitions are executed with large horizons while the grasping and uprighting phases shrink to smaller values to handle the precise alignment with the thin circular rim and the controlled rotation. The pick-and-place tape task in Fig. 4(c) follows a more standard pattern, in which the horizon shrinks only at the two critical moments of grasping and releasing, while remaining large during the rest of the trajectory, leading to a more compact overall execution.

These comparisons demonstrate that the proposed adaptive horizon mechanism is able to allocate control granularity according to the underlying task structure, executing efficiently in simple settings and operating with finer precision in more complex ones, all without task-specific tuning.

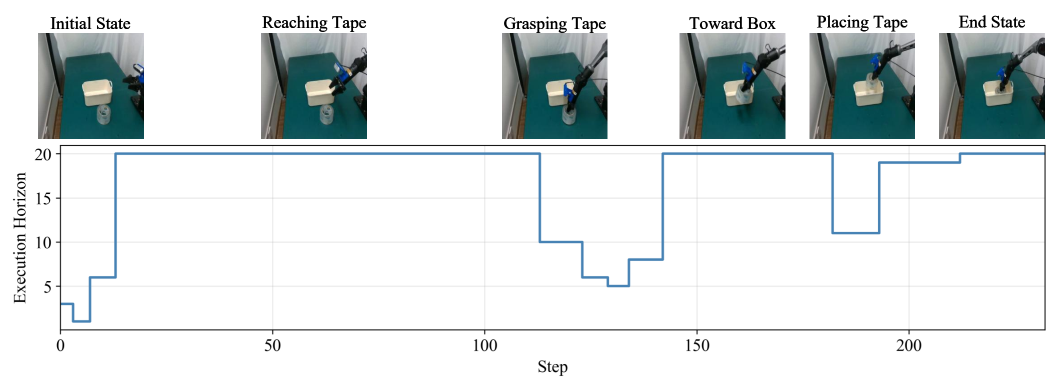
Failure case analysis. We analyze two representative failure cases from real-world rollouts. As shown in Fig. 5(a), in the *hangmug* task, the mug handle fails to align with the hook during the placing phase. This task requires sub-centimeter alignment, and small residual errors in the predicted end-effector



(a) Pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate



(b) Flip the mug



(c) Cick up the tape and place it in the box

Figure 4: Visualization of the execution horizon across different tasks.

pose, though tolerable earlier in the trajectory, become unrecoverable at the moment of contact. A^3 improves the reliability of committed action prefixes but does not enhance the spatial precision of the underlying policy, and is therefore insufficient to resolve such fine-grained contact-rich failures.

As shown in Fig. 5(b), in the *flipmug* task, the gripper completely occludes the inverted mug rim from the top-down camera view during approach, leaving the policy without effective visual feedback at the critical closing moment and ultimately leading to a failed grasp. Since A^3 verifies action consistency under the model's existing observations, it cannot compensate for missing visual information caused by viewpoint occlusion. Overall, these cases indicate that adaptive execution commitment alone is insufficient when failures originate from perceptual coverage or control precision limits of the underlying policy.

(a) Failure case 1

(b) Failure case 2

Figure 5: Two representative failure cases. **(a)** Misalignment between the mug handle and the hook in the *hang mug* task. **(b)** Self-occlusion of the inverted mug rim by the gripper in the *flip mug* task.

6 Conclusion and Future Work

In this work, we identify the determination of execution commitment as a principled yet under-explored inference problem in multi-step VLA systems, and reformulate horizon selection as a state-conditioned prefix verification problem. Instead of relying on predefined execution windows or heuristic uncertainty thresholds, A^3 treats predicted action chunks as speculative drafts and adaptively determines the executable horizon through structured self-verification, without auxiliary modules or task-specific retraining. By combining trajectory-level consensus estimation with dual hierarchical verification constraints, *i.e.,* consensus-ordered conditional invariance and prefix-closed sequential consistency, A^3 grounds execution decisions in both internal model logic and physical rollout feasibility. Extensive experiments across diverse VLA backbones and benchmarks demonstrate that A^3 eliminates the need for manual horizon tuning while consistently matching or outperforming exhaustively grid-searched fixed horizons, achieving a superior trade-off between execution robustness and inference efficiency.

Future Work. While A^3 demonstrates the viability of self-speculative prefix verification for adaptive execution commitment, several directions remain open. First, replacing group sampling with lightweight learned reliability estimators could reduce inference overhead without sacrificing verification quality. Second, incorporating explicit world models or learned state transition predictors would extend verification beyond model self-consistency, yielding stronger guarantees under distribution shift. Finally, relaxing the prefix-closed constraint to support non-contiguous or adaptively reordered commitment structures may further improve robustness in long-horizon manipulation tasks.

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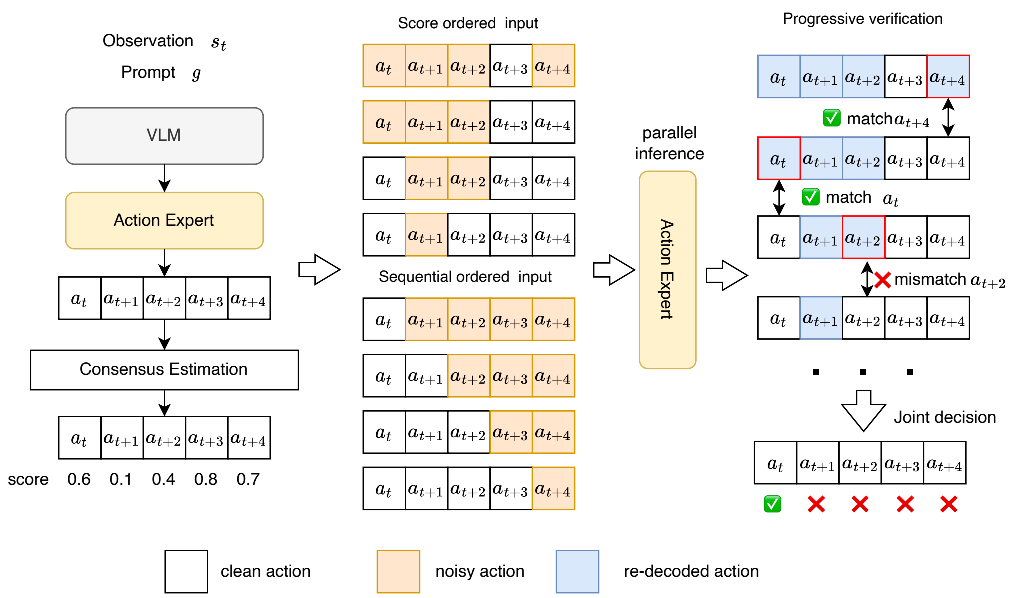

Figure 6: Implementation of dual verification tree.

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A Implementation Details

Implementation of the verification tree. As shown in Figure 6, following the self-speculative decoding paradigm of SSD [15], we implement the dual hierarchical verification as a batched tree over candidate action prefixes. For a prediction horizon of H actions, the two verification constraints—consensus-ordered conditional invariance and prefix-closed sequential consistency—each requires H re-decoding queries, yielding $2H$ total inpainting inputs. Rather than executing these sequentially, we construct all $2H$ queries in parallel: for each action position i, the corresponding context actions (either $\mathcal{C}_i^{\text{inv}}$ or $\mathcal{C}_i^{\text{seq}}$) are fixed at their draft values via flow matching inpainting, while position i is freely denoised. All $2H$ inpainting inputs are stacked into a single batch and processed in one forward pass through the action expert, after which the re-decoded outputs are compared against the original draft using the acceptance threshold δ. This design ensures that the entire verification stage adds exactly one forward call regardless of horizon length H, keeping the per-step overhead constant and amortizable over the committed action prefix.

Algorithm. The algorithm of our method can be found in Alg. 1.

Algorithm 1 Adaptive Action Acceptance (A^3)

Require: VLA policy π_θ, current state s_t, prediction horizon H, number of samples K, acceptance threshold δ, alignment window w, temperature τ, kinematic integrator $\mathcal{G}$

Ensure: Committed action prefix $\mathbf{A}_t[1 : l^*]$

Stage 1: Mode-aware Trajectory Consensus Scoring

Generate K candidate chunks via parallel noise injection: $\{\mathbf{A}_t^{(k)}\}_{k=1}^{K} \leftarrow \pi_\theta(s_t,\, \epsilon^{(k)}), \quad \epsilon^{(k)} \sim \mathcal{N}(0, \sigma^2 I)$
Compute induced trajectories $\{T^{(k)}\}_{k=1}^{K}$ via kinematic integrator $\mathcal{G}$
for each pair (k, k') **do**
  $D(k, k') = \sum_j \min_{|\Delta| \le w} \|\hat{s}_{t+j}^{(k)} - \hat{s}_{t+j+\Delta}^{(k')}\|_2^2$
end for
Cluster $\{T^{(k)}\}$ under $D(\cdot, \cdot)$; score each cluster c: $S_c = p_c \cdot \exp(-\mathrm{Disp}(c)/\tau)$
$c^* \leftarrow \arg\max_c S_c$; $k^* \leftarrow \arg\min_{k \in c^*} \sum_{k' \in c^*} D(k, k')$ ▷ dominant mode & medoid draft
Compute per-step consensus score $\tilde{R}_j$ for $a_{t+j} \in \mathbf{A}_t^{(k^*)}$ via Eq. (5)

Stage 2: Dual Hierarchical Verification via Flow Matching Inpainting

Sort action indices by consensus (descending): $\pi_R \leftarrow \mathrm{argsort}(\{\tilde{R}_j\}_{j=1}^{H}, \downarrow)$
Construct batched inpainting inputs $\mathcal{B} \leftarrow \emptyset$
for each action index $i \in \{1, \ldots, H\}$ **do**
  $\mathcal{C}_i^{\mathrm{inv}} \leftarrow \{j \mid \tilde{R}_j > \tilde{R}_i\}$ ▷ higher-consensus positions as fixed context
  $\mathcal{C}_i^{\mathrm{seq}} \leftarrow \{j \mid j < i\}$ ▷ sequential prefix positions as fixed context
  Construct inpainting mask m_i^{inv}: fix positions $\mathcal{C}_i^{\mathrm{inv}}$ to $\mathbf{A}_t^{(k^*)}$, denoise position i
  Construct inpainting mask m_i^{seq}: fix positions $\mathcal{C}_i^{\mathrm{seq}}$ to $\mathbf{A}_t^{(k^*)}$, denoise position i
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(m_i^{\mathrm{inv}},\, m_i^{\mathrm{seq}})\}$
end for
Execute single batched flow matching denoising pass over $\mathcal{B}$: $\{\tilde{a}_i^{\mathrm{inv}},\, \tilde{a}_i^{\mathrm{seq}}\}_{i=1}^{H} \leftarrow \pi_\theta(s_t \mid \mathcal{B})$
for each action index $i \in \{1, \ldots, H\}$ **do**
  $V_i^{\mathrm{inv}} \leftarrow \mathbf{1}\big[\, |\tilde{a}_i^{\mathrm{inv}} - a_{t+i}| < \delta \,\big]$ ▷ conditional invariance check
  $V_i^{\mathrm{seq}} \leftarrow \mathbf{1}\big[\, |\tilde{a}_i^{\mathrm{seq}} - a_{t+i}| < \delta \,\big]$ ▷ sequential consistency check
end for

Stage 3: Prefix-closed Horizon Determination

$l^* \leftarrow \max\{l \in \{1, \ldots, H\} \mid \forall\, i \le l : \; V_i^{\mathrm{inv}} = 1 \;\wedge\; V_i^{\mathrm{seq}} = 1\}$
return $\mathbf{A}_t[1 : l^*] = \{a_t, \ldots, a_{t+l^*-1}\}$